# Vasudha Pai Karkala

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## Research Experience

**Research Intern**, Associated Laboratory for Green Chemistry (LAQV/REQUIMTE), Portugal — May 2026 – Present

- Contributed to developing cheminformatics workflows with RDKit to analyze ChEMBL bioactivity data and identify potential IL-6 inhibitors.
- Computer-aided drug design (CADD) and cheminformatics to accelerate immunotherapy-focused drug discovery.
- Developed QSAR Models to optimize CAR-T cell therapy.

**Summer Research Intern**, Associated Laboratory for Green Chemistry (LAQV/REQUIMTE), Portugal — June 2026 – July 2026

- Worked on the synthesis and development of biocompatible ionic liquids as co-solvents to improve the solubility of BCS Class IV drugs in physiological media.
- Characterized and evaluated the ionic liquids with Sulfamethoxazole and Acetazolamide.

**Young Research Fellow in Computational Chemistry**, I-hub Data, International Institute of Information and Technology (IIIT-H), India — Aug 2024 – Aug 2025

- Analyzed trajectories for the PLAS 20K and PLAS HES-5K datasets to extract and interpret dynamic patterns of Protein-Ligand Complexes from Steered Molecular Dynamics(SMD).
- Conducted Linear Interaction Energy (LIE) analysis for Protein-Ligand complexes to evaluate binding affinities.
- Calculated the water number between the protein-ligand interactions.

**Project Intern**, Dr. Reddy's Institute of Life Sciences (DRILS), India — Jan 2024 – July 2024

- Worked on structurally characterizing the Gain of Function mutation of p53 and understanding its allosteric interactions with proteins.
- Employed molecular dynamics simulations using GROMACS to probe allosteric behaviour in loss of function and gain of function mutants.
- Used tools like MD-TASK to understand the single-nucleotide variants present in the mutants.

## Poster and oral Presentations

**CS-3 Strasbourg Summer School** — June 2026

- Presented a poster on "Integrating Morgan Fingerprints and Physicochemical Descriptors for Solubility Prediction" at the Strasbourg Summer School.
- Developed hybrid datasets to predict the Aqueous Solubility using Random Forest and XG-Boost.

**Tech For Sustainable Futures** — March 2025

- Presented a poster on "From Data to Discovery:PLAS-20K and AP Dataset" at R and D Showcase 2025- IIIT-H.

## Honors and Achievements

**Erasmus Mundus Scholarship:** Recipient of the Erasmus Mundus Joint Master's Scholarship (ChEMoinformatics+).

**Achiever's Scholarship:** Secured the prestigious Achiever's scholarship after achieving first place in the Bioinformatics Master's program.

**Second place in Biotech Hackathon:** Won Second prize in the Manipal Biotech Hackathon conducted by the Manipal-GOK Bioincubator.

## Education

**Erasmus Mundus Joint Master's in Chemoinformatics+, Portugal and France** Sept 2025 – Present
MS in Bio-Organic Chemistry (Lisbon) and Cheminformatics (Strasbourg)

- **Coursework:** Asymmetric Organic Chemistry, Python, Medicinal Chemistry, Analytical Chemistry, Synthetic Organic Chemistry and Computational Chemistry

**Manipal Academy of Higher Education, India**, MS in Bioinformatics Aug 2022 – July 2024

- **Coursework:** Biocomputing, Structural Bioinformatics, Genomics and Proteomics, R Programming.

**Osmania University, India**, BS in Biotechnology, Microbiology and Chemistry June 2018 – May 2021

- **Coursework:** Cellular and Molecular Biology, Biochemistry, Industrial Biotechnology, Plant and animal tissue culture

## Volunteering

**Volunteering Service Organization(VSO, MAHE)**

- Actively participated in VSO MAHE, contributing to community-driven projects aimed by engaging with children with special needs, providing English language instruction and fostering an inclusive learning environment.

**National Service Scheme (NSS)**

- Participated in the National Service Scheme (NSS), contributing to community welfare programs and outreach initiatives focused on social development, public health awareness, and environmental sustainability.

## Languages and Technical SKills

**Programming and Computational Languages:** Python, Perl, R, HTML.
**Data Science and Cheminformatics Libraries:** Numpy, Pandas, Scikit-learn, Matplotlib, RDKit, Openbabel, Gaussview, Reinvent4